SKILL GOBANG

CS 307 BK project

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Background & Motivation

Previous course project: A team developed a command-line Gobang (Five-in-a-Row) implementation

Strengths of prior work:
- Solid core game logic implementation
- Functional win-detection algorithm

Identified improvement opportunities:

1. Replace command-line interface with intuitive GUI for better user experience

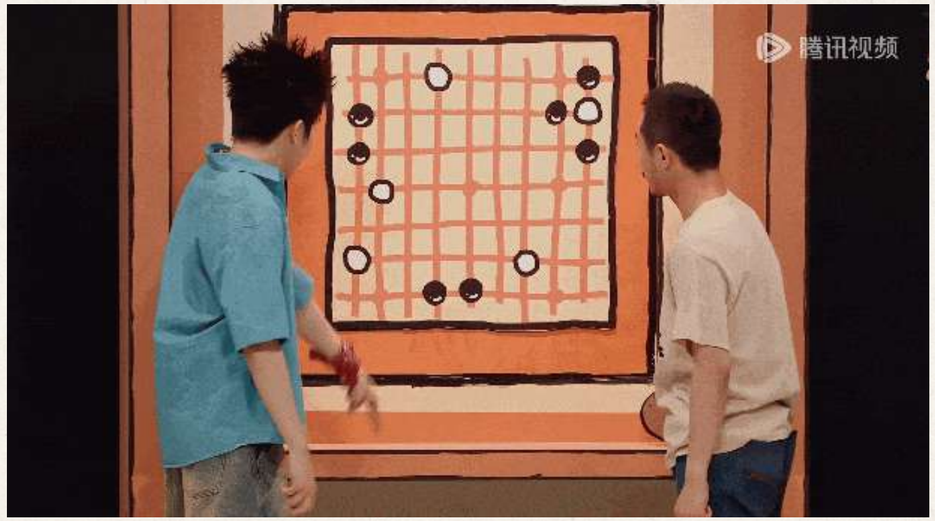
2. Integrate intelligent AI opponent with adjustable difficulty levels

3. Enhance gameplay engagement beyond traditional rules

Beyond Five-in-a-Row: The Core Concept

It’s not just about where you place your piece, it’s about how you break the rules.

- **The Problem:** Traditional Gobang can feel “solved” or repetitive at high levels.
- **The Solution:** Skills introduce variables that force players to adapt their strategy in real-time.



技能五子棋*(Skill Gobang)*, *2025*, *Zhang et al.*

Innovation & Technology Stack

🗺️ Board Manipulation: Removing/Moving pieces.
🔄 Rule Alteration: Swapping turns or double-placement.

- **Frontend**: React.js + TypeScript
- **Backend**: Node.js + Socket.IO (Real-time)
- **AI Intelligence**: LLM API (GPT) for skill suggestions

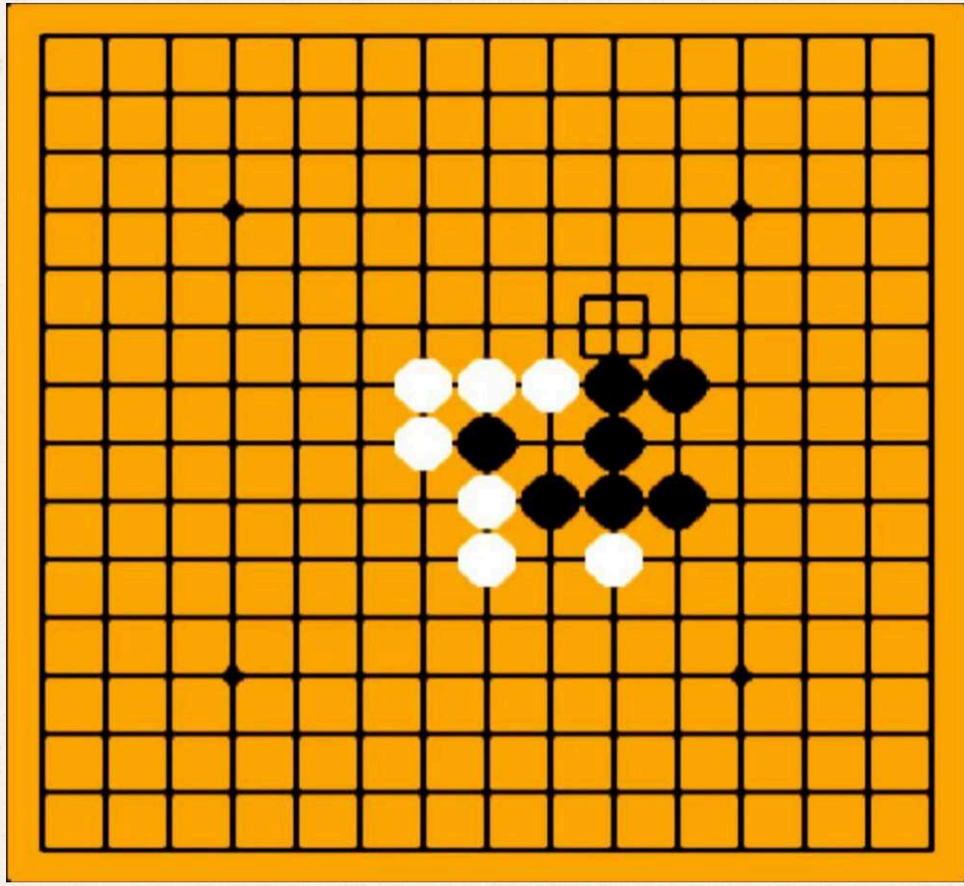
Game Design Mechanisms

Basic Rules

Players take turns placing pieces.

Each turn, place one piece on an empty space on the board.

Five pieces in a row → You win!



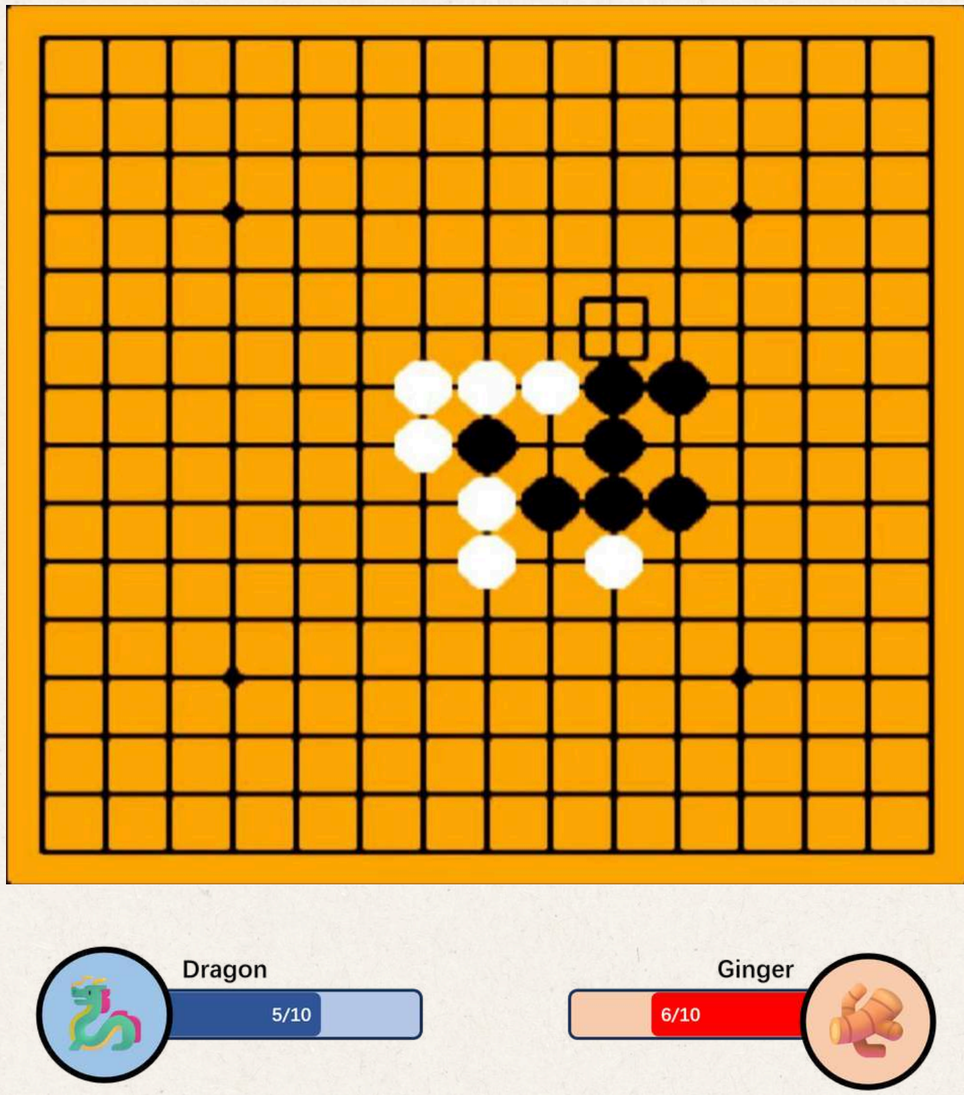
Game Design Mechanisms

Energy

At the start of a player's turn, automatically gain one energy.

When a player aligns three pieces in a row, gain one energy.

So what?

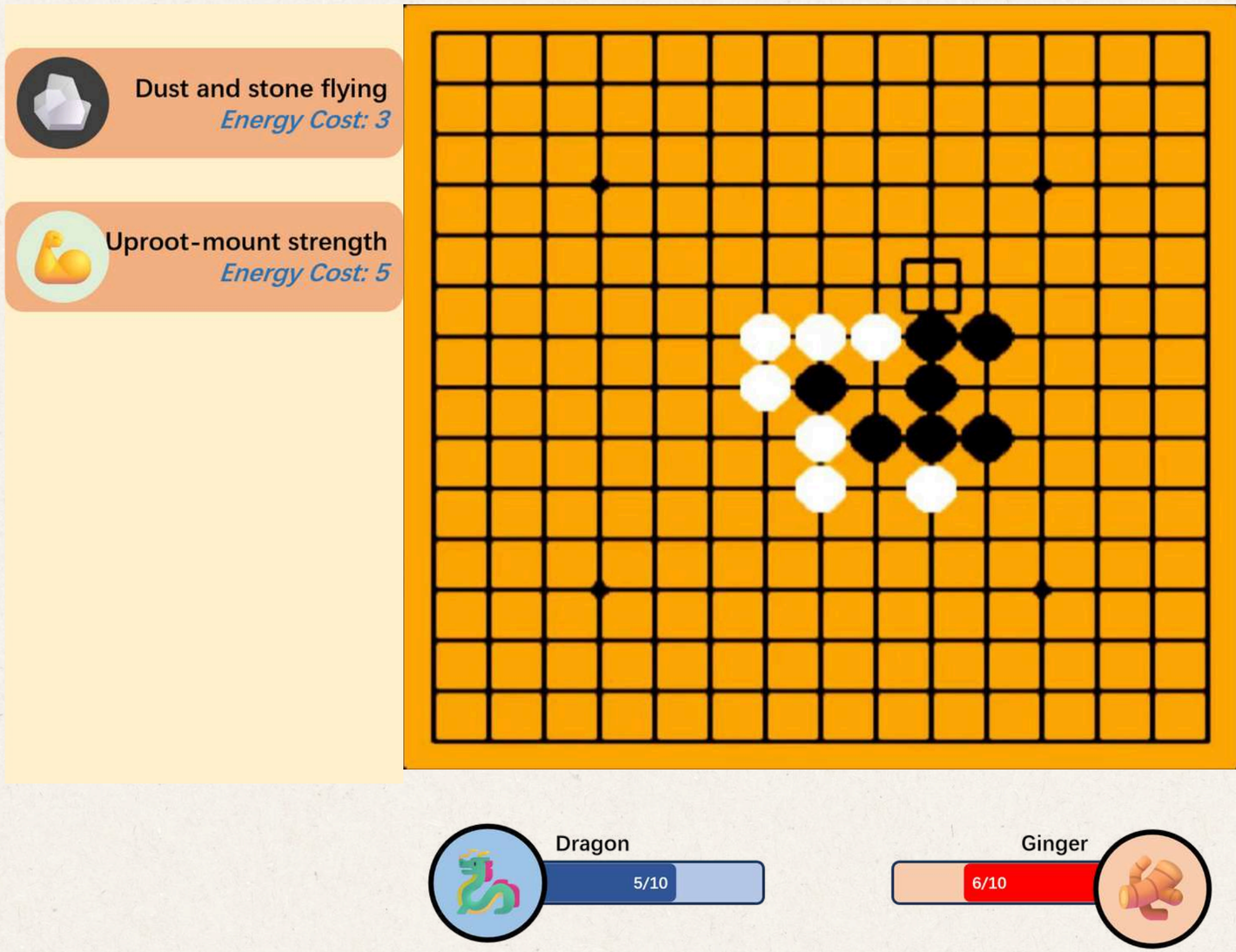


Game Design Mechanisms

Skill

On their turn, players may choose to pass their move and use a skill.

Each skill will consume a certain amount of energy.



Some Skills

Dust and stone flying

Cost: 3

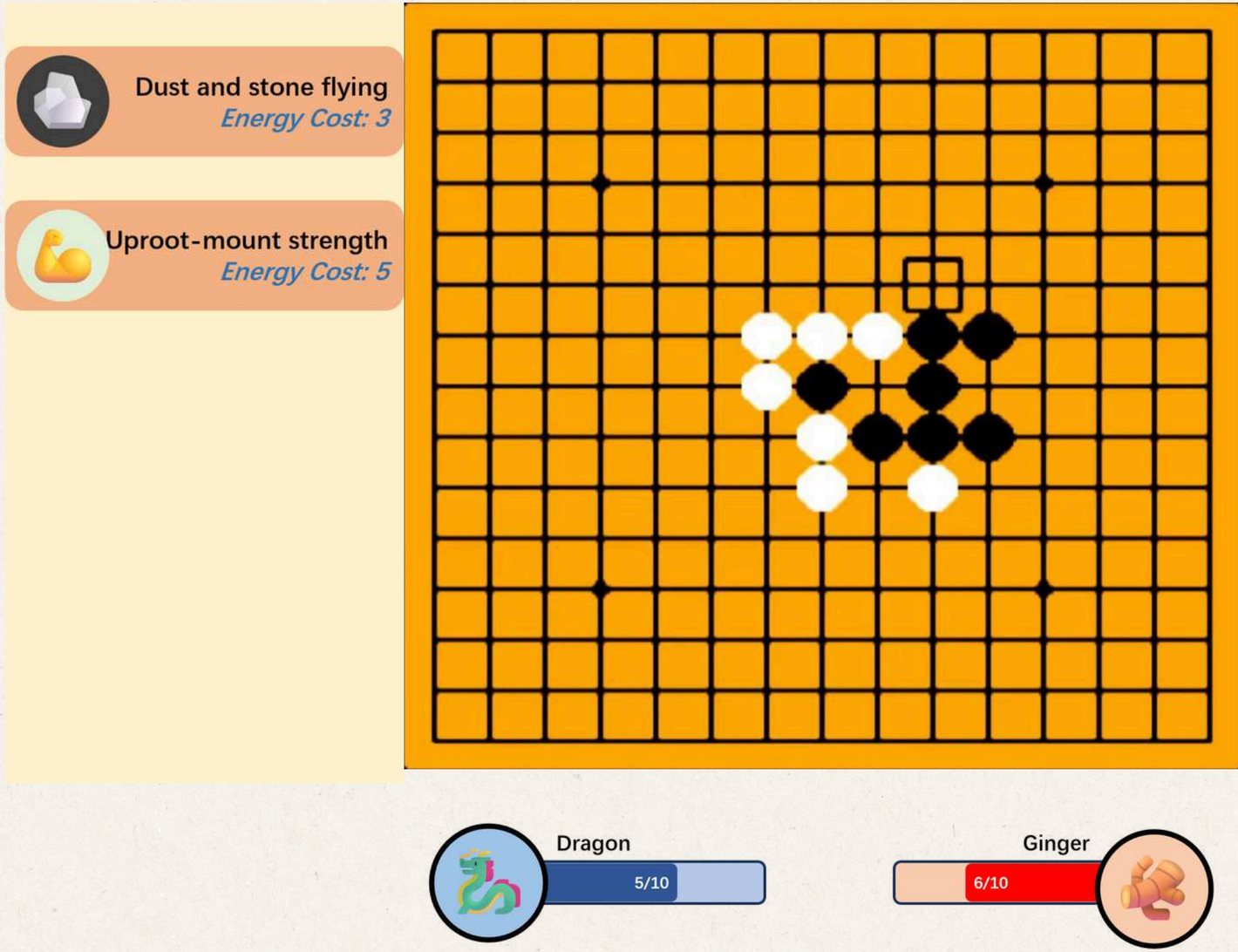
Remove one of your opponent's pieces from the board.

Uproot-mount strength

Cost: 5

After use, the following outcomes may occur:

- 50% chance: Win immediately
- 30% chance: Gain one additional turn
- 20% chance: Nothing happens; opponent's turn



Project Timeline

Week 1 – Core System

- GUI with board visualization
- Real-time multiplayer infrastructure
- API design for AI integration
- Deliverable: Human vs. Human playable version

Week 2 – AI & Refinement

- Skill-aware AI with difficulty levels
- Rule validation + fallback logic
- Playtesting and balance updates
- Deliverable: Complete game with AI opponent

AI Usage & Human Contribution

Human Role

- Designed gameplay mechanics and skills
- Built frontend and backend systems
- Defined rules and validation
- Tested and refined user experience

AI Role

- Strategic move suggestions
- Skill recommendations
- Adaptive opponent behavior
- Development support (debugging/code help)

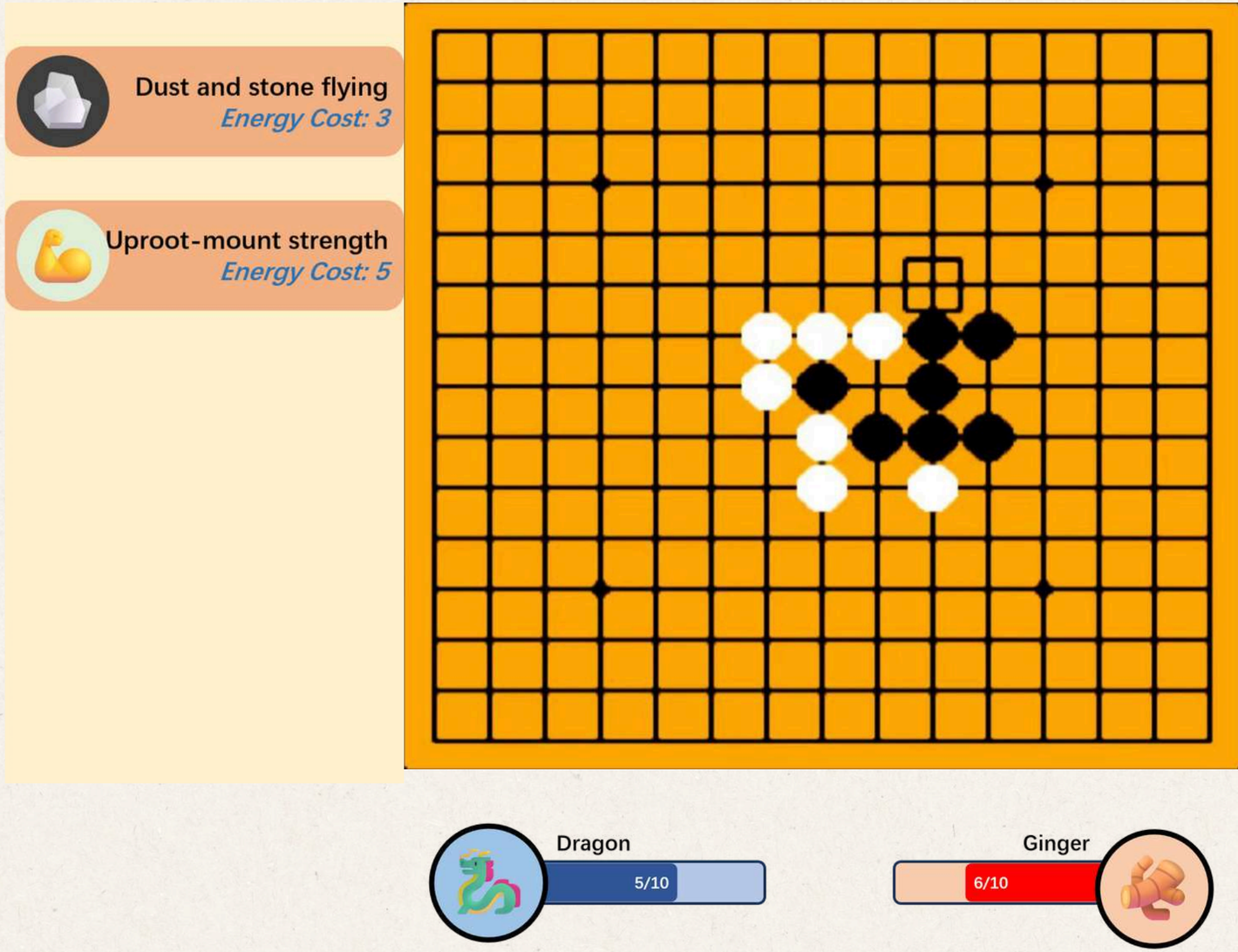
Principle: AI assists. Humans control design and decisions.

Small Quiz

As a player, what matters most in skill-based Gobang?

A. Masterful Gobang technique

B. Proficient application of skills

C. Winning

D. Enjoyment